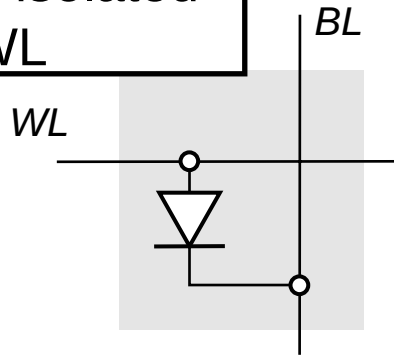


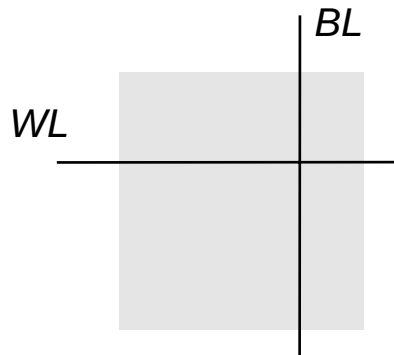
Read-Only Memory Cells

BL not isolated from WL

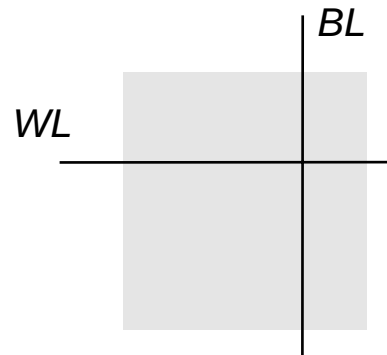
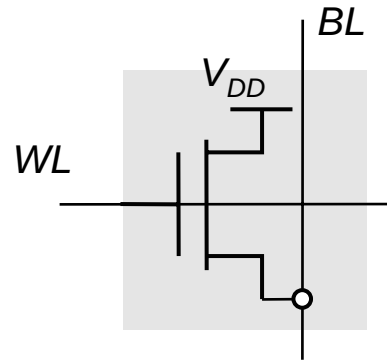
1



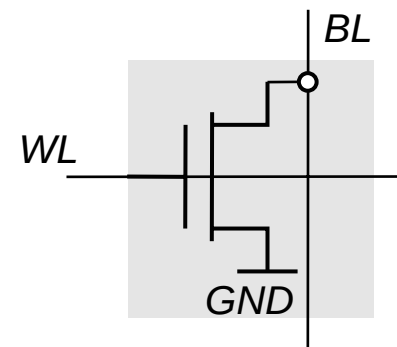
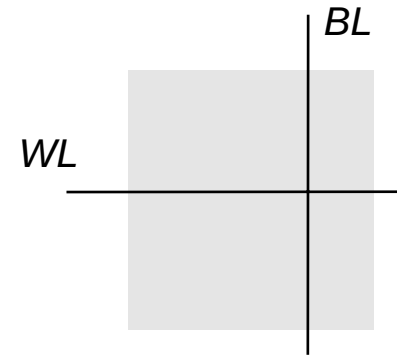
0



Diode ROM

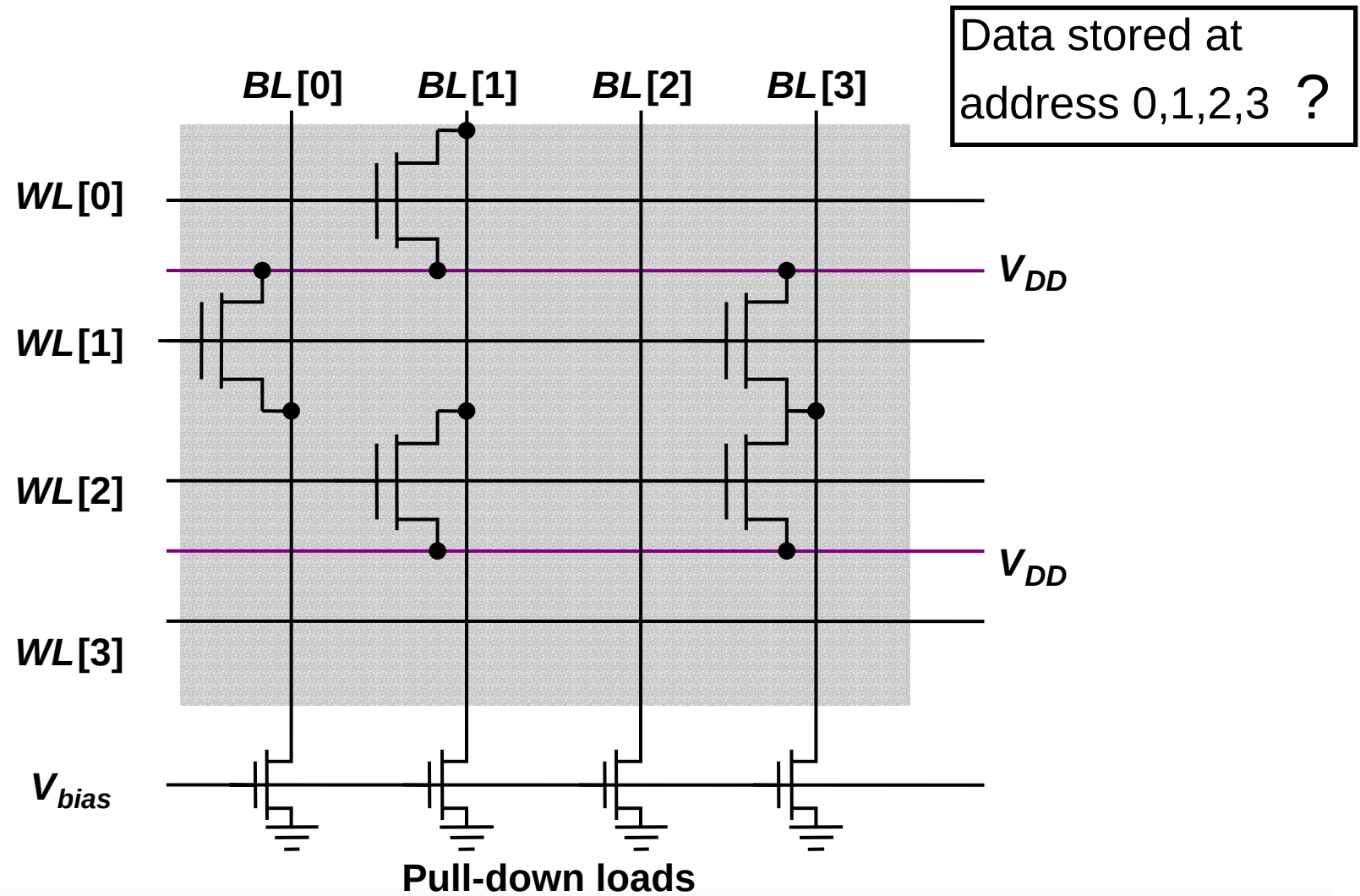


MOS ROM 1



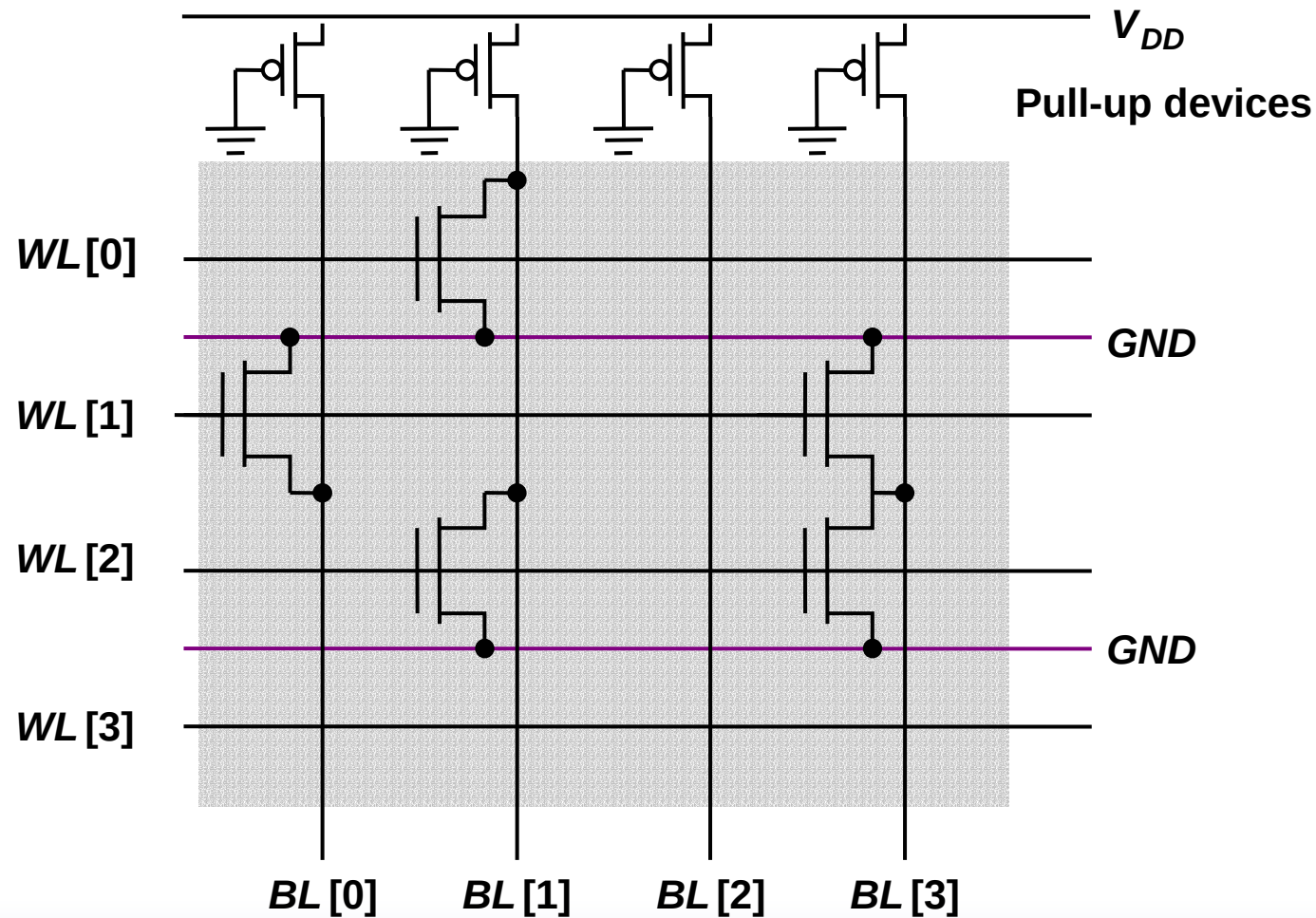
MOS ROM 2

MOS OR ROM



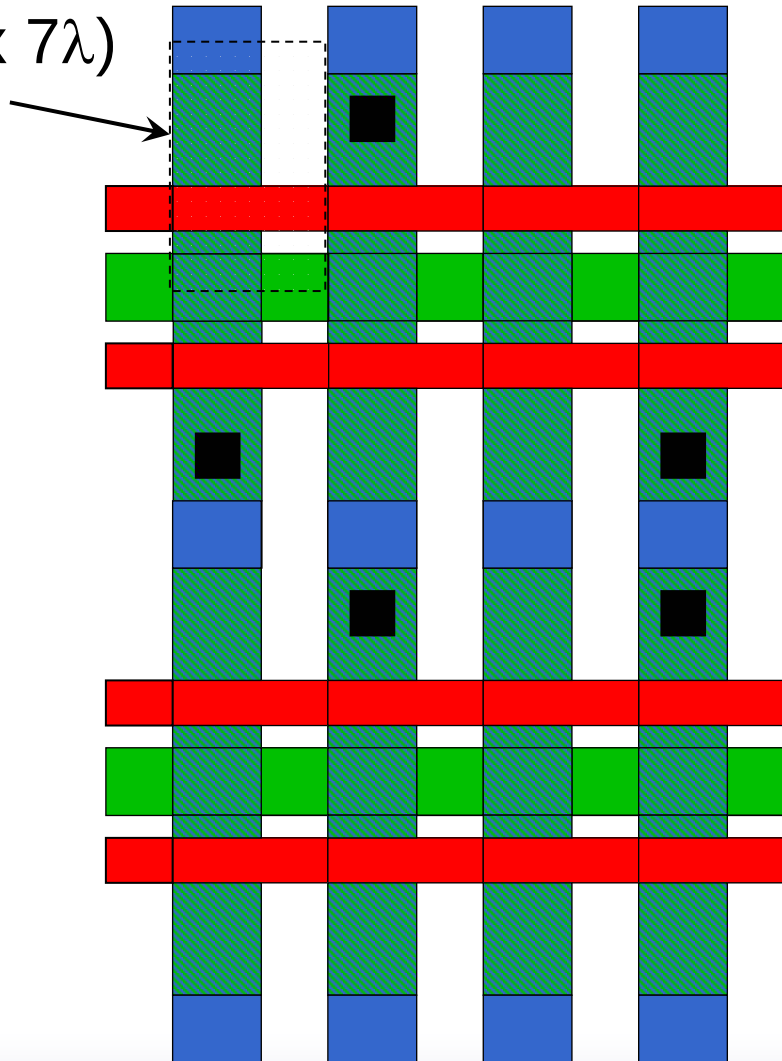
MOS NOR ROM

Data stored at
address 0,1,2,3 ?



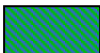
MOS NOR ROM Layout

Cell ($11\lambda \times 7\lambda$)



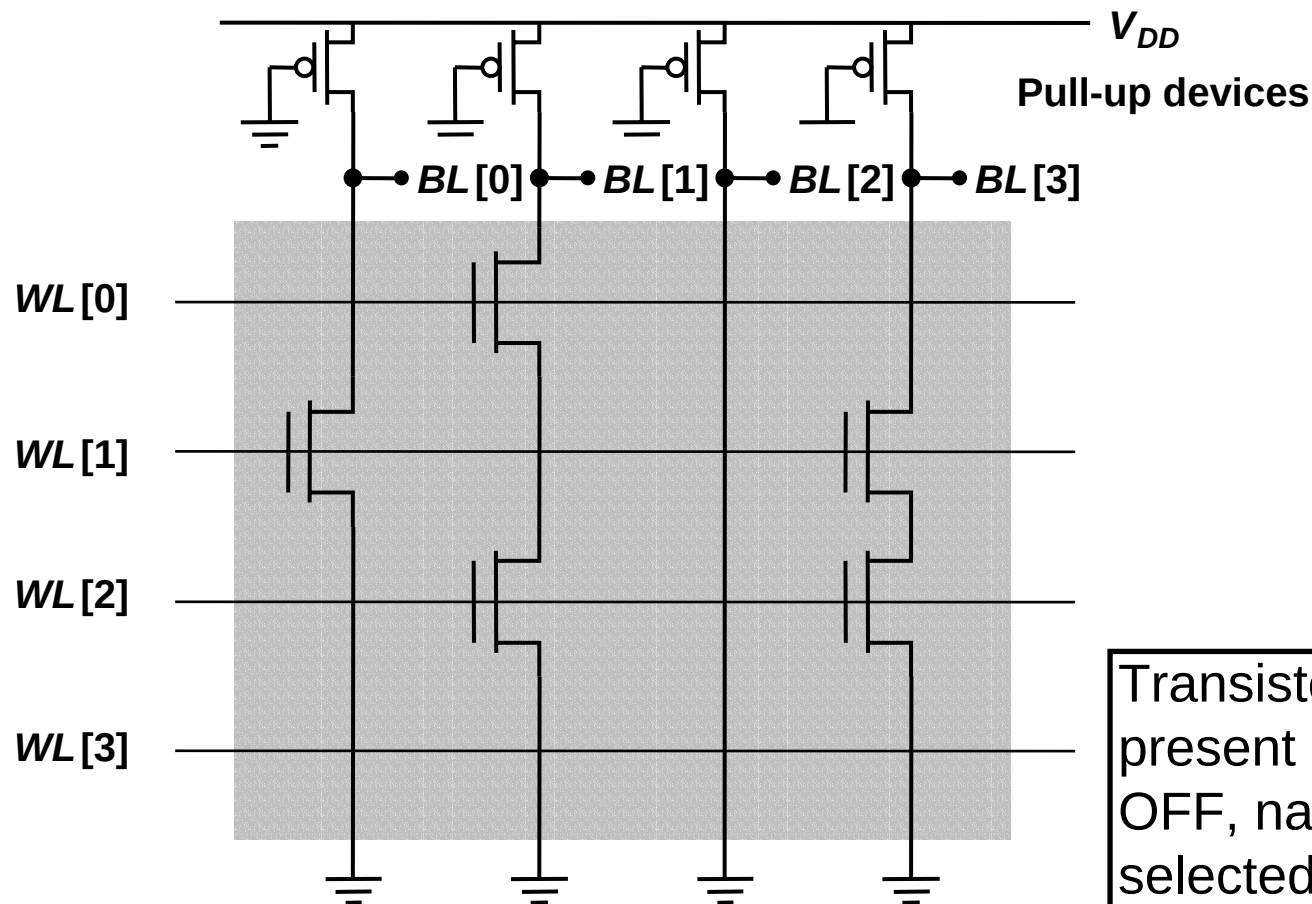
A large part of the cell is devoted to contacts to bitline and GND

Programmming using the Contact Layer Only

-  Polysilicon
-  Metal1
-  Diffusion
-  Metal1 on Diffusion

MOS NAND ROM

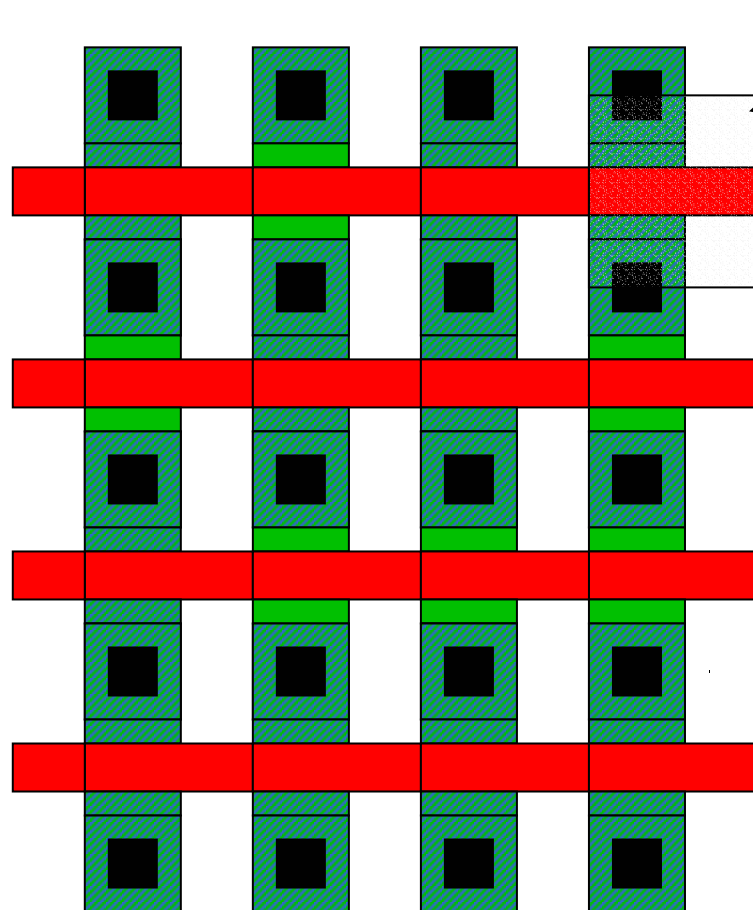
Data stored at
address 0,1,2,3 ?



Transistors that are not present cannot be switched OFF, namely they cannot be selected

All word lines high by default with exception of selected row

MOS NAND ROM Layout



Cell ($8\lambda \times 7\lambda$)

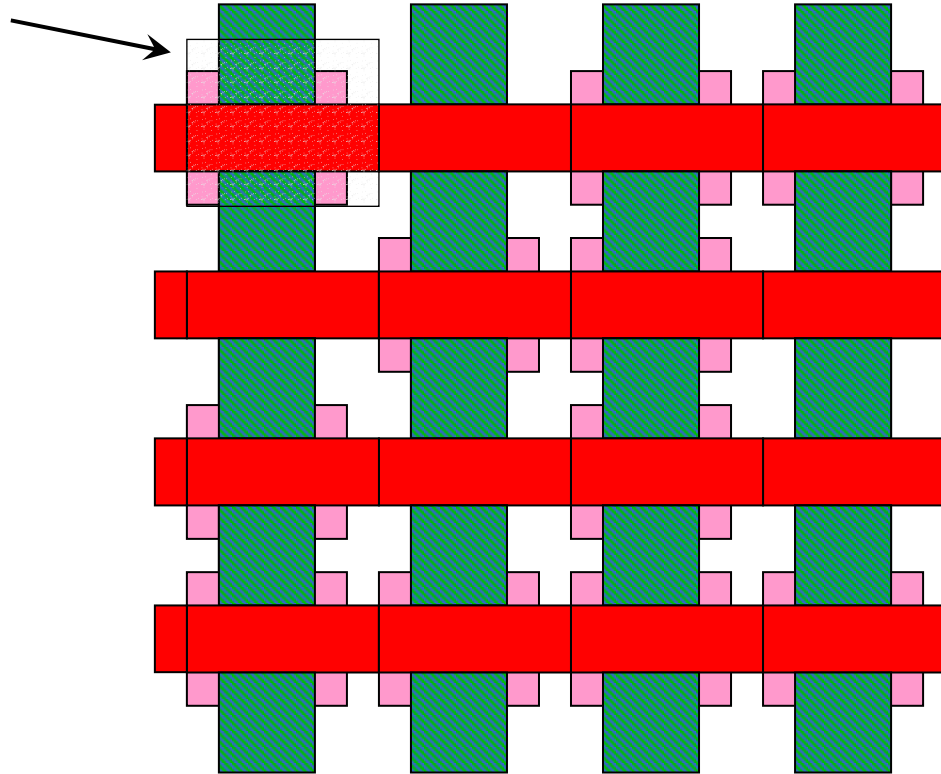
**Programmming using
the Metal-1 Layer Only**

No contact to VDD or GND necessary;
drastically reduced cell size
Loss in performance compared to NOR ROM

-  Polysilicon
-  Diffusion
-  Metal1 on Diffusion

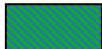
NAND ROM Layout

Cell ($5\lambda \times 6\lambda$)



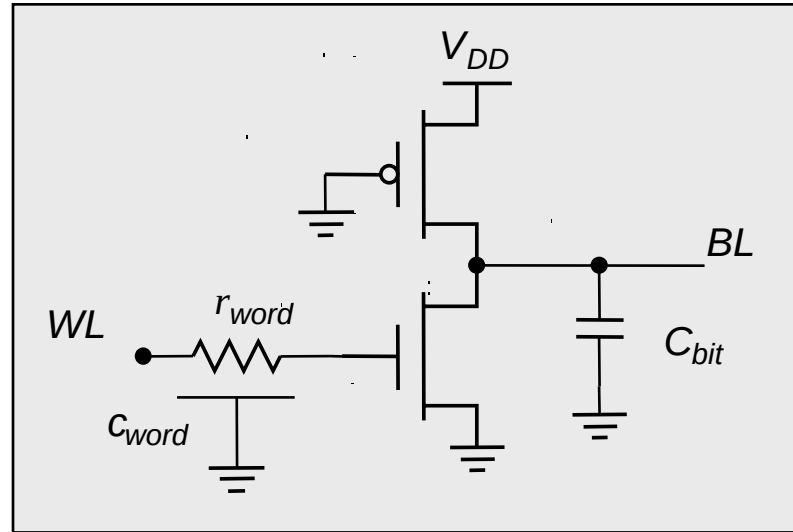
**Programmming using
Implants Only**

n-type implant turns the device into a depletion transistor that is always ON regardless of the word line voltage

-  Polysilicon
-  Threshold-altering implant
-  Metal1 on Diffusion

Equivalent Transient Model for MOS NOR ROM

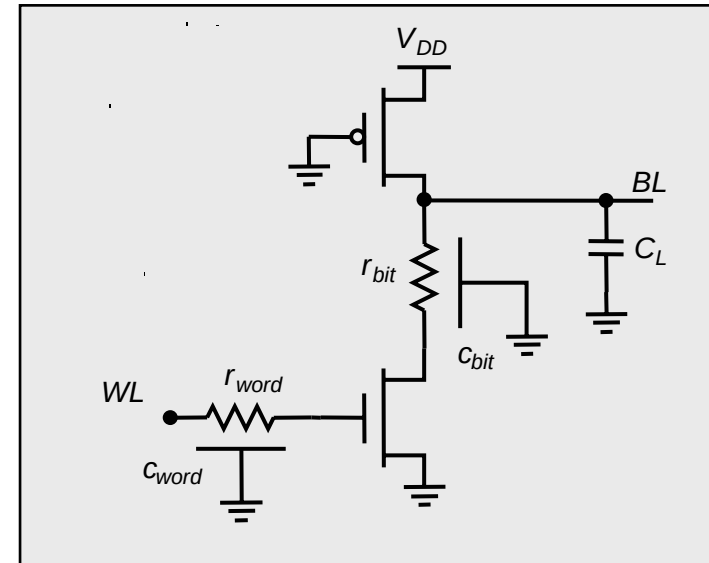
Model for NOR ROM



- ❑ Word line parasitics
 - Wire capacitance and gate capacitance
 - Wire resistance (polysilicon)
- ❑ Bit line parasitics
 - Resistance not dominant (metal)
 - Drain and Gate-Drain capacitance

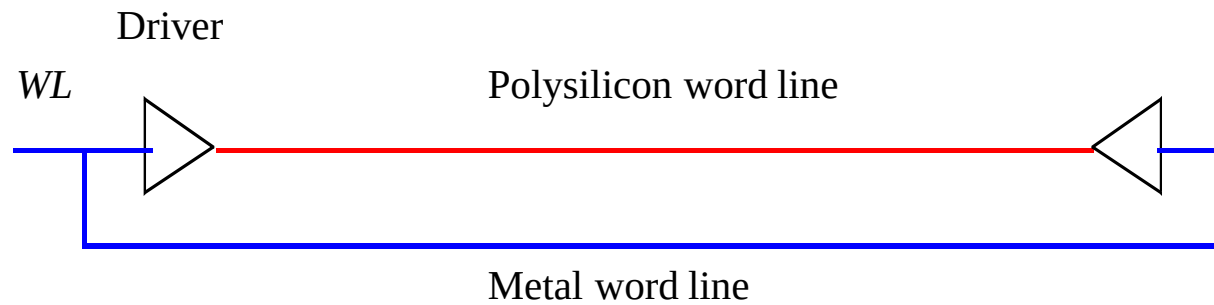
Equivalent Transient Model for MOS NAND ROM

Model for NAND ROM

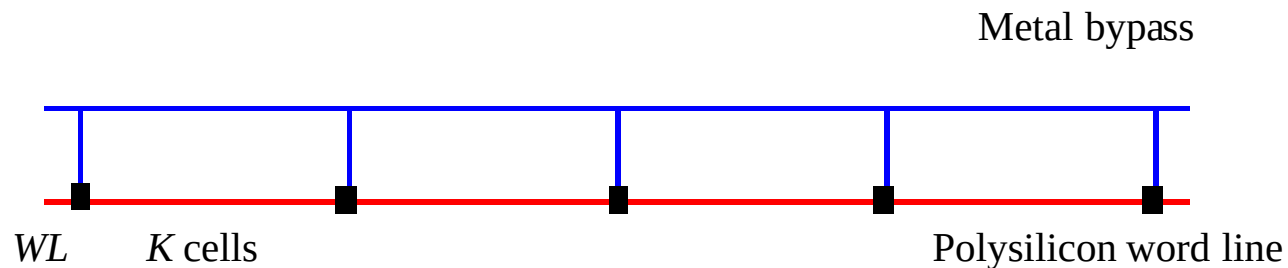


- ❑ Word line parasitics
 - Similar to NOR ROM
- ❑ Bit line parasitics
 - Resistance of cascaded transistors dominates
 - Drain/Source and complete gate capacitance

Decreasing Word Line Delay



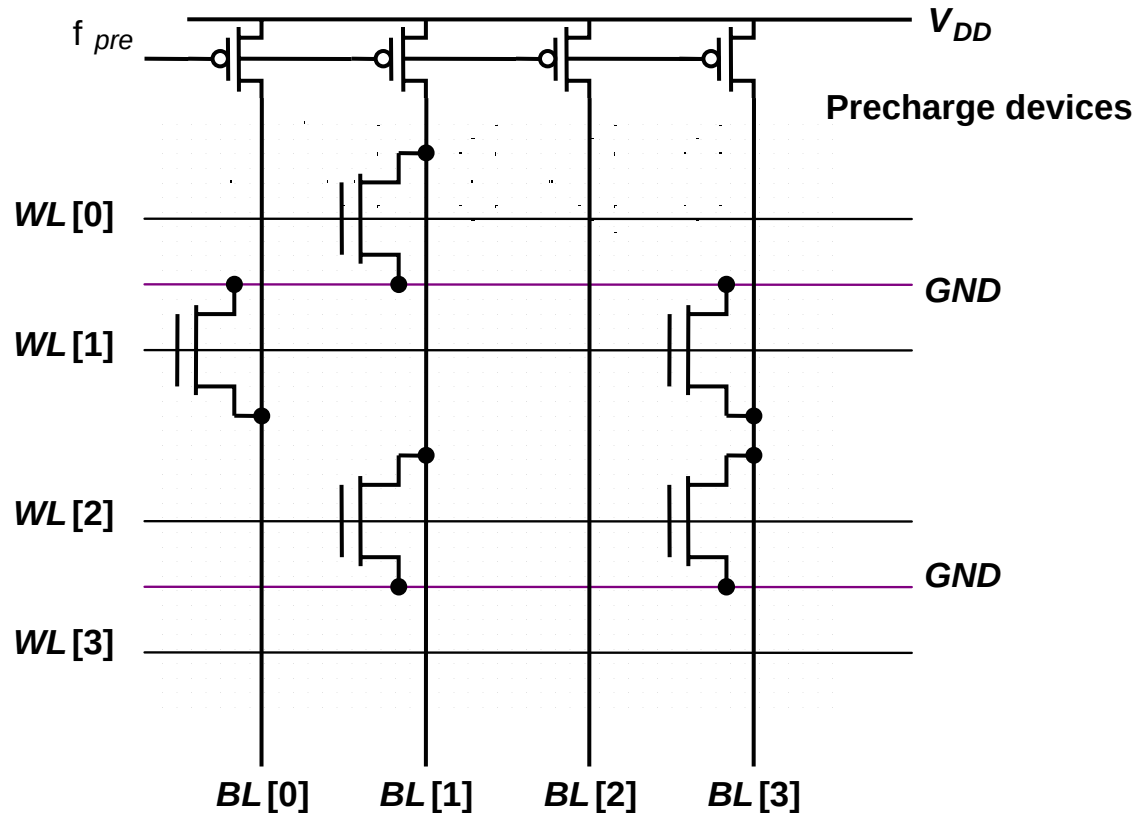
(a) Driving the word line from both sides



(b) Using a metal bypass

(c) Use silicides

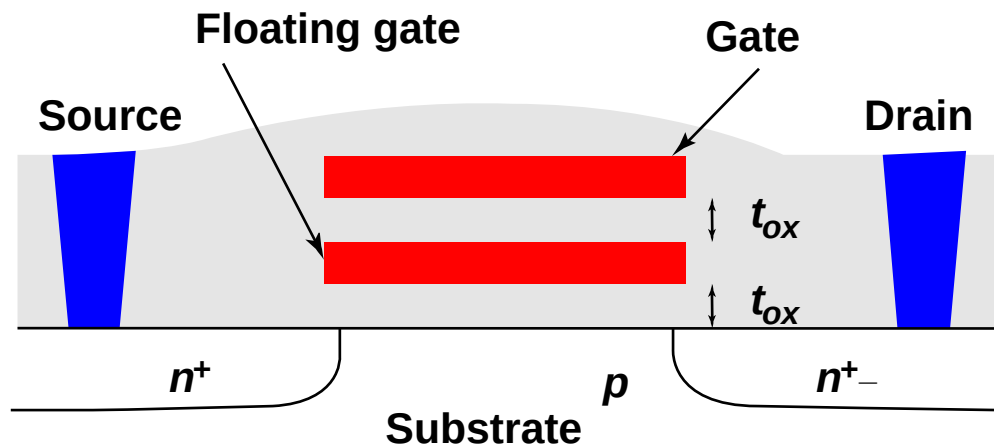
Precharged MOS NOR ROM



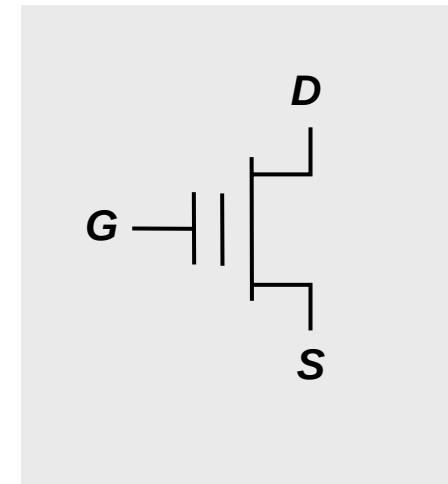
PMOS precharge device can be made as large as necessary, but clock driver becomes harder to design.

Non-Volatile Memories

The Floating-gate transistor (FAMOS)

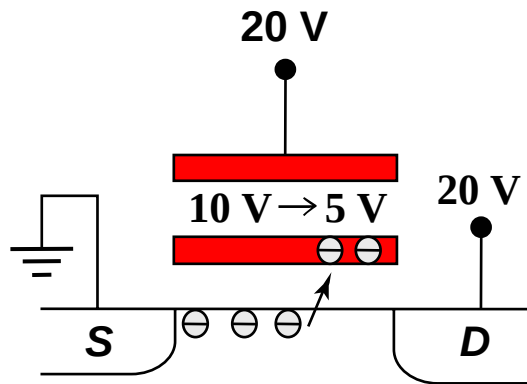


Device cross-section

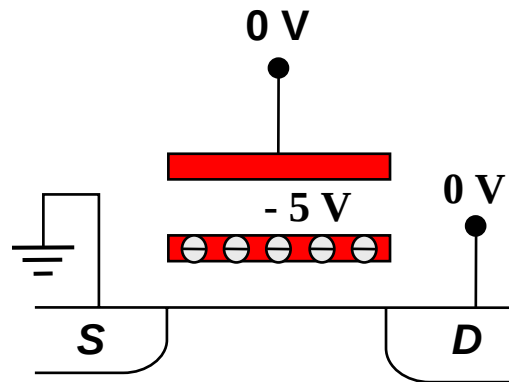


Schematic symbol

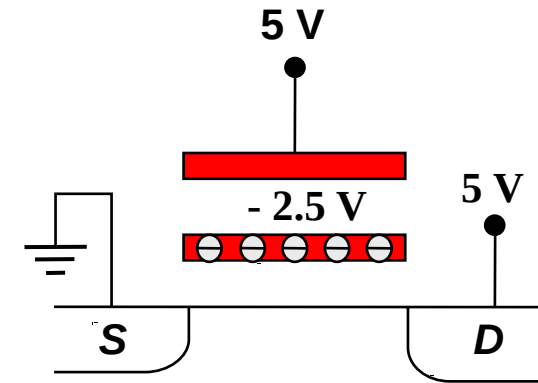
Floating-Gate Transistor Programming



Hot-carrier injection

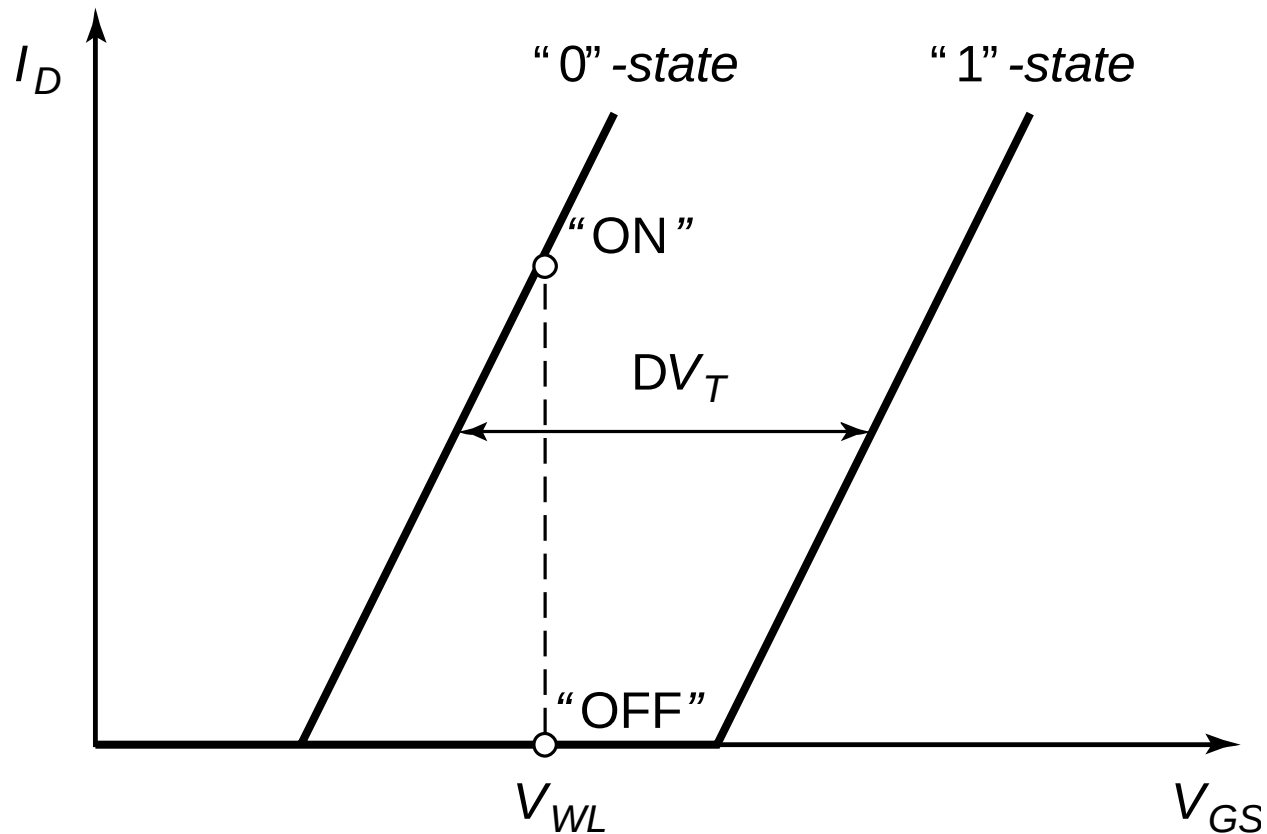


Removing programming voltage leaves charge trapped

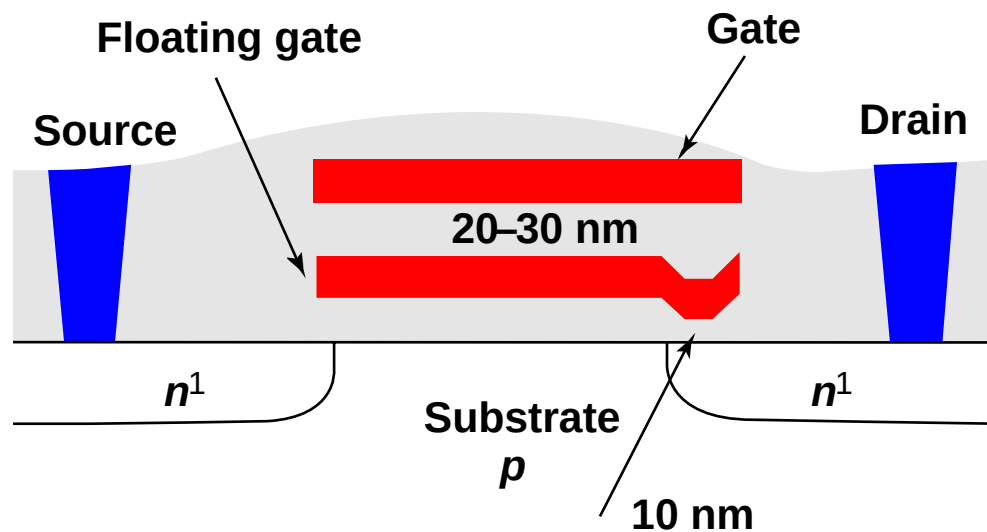


Programming results in higher V_T .

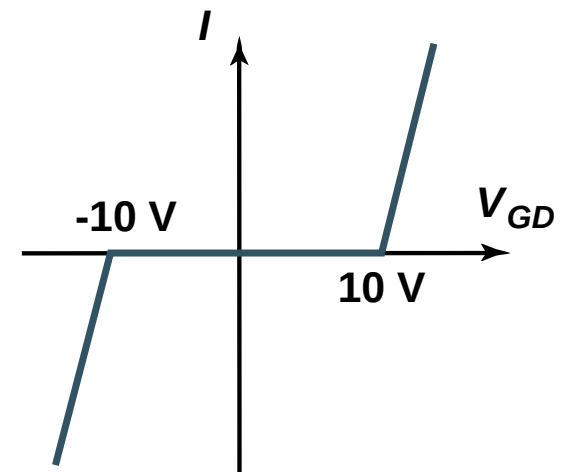
A “Programmable-Threshold” Transistor



FLOTOX EEPROM

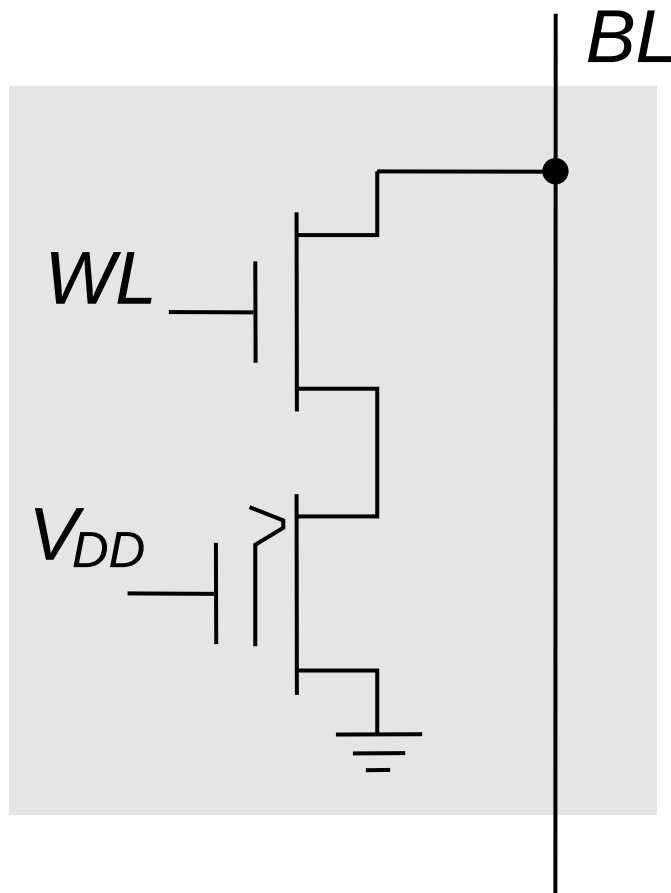


FLOTOX transistor



**Fowler-Nordheim
 I - V characteristic**

EEPROM Cell

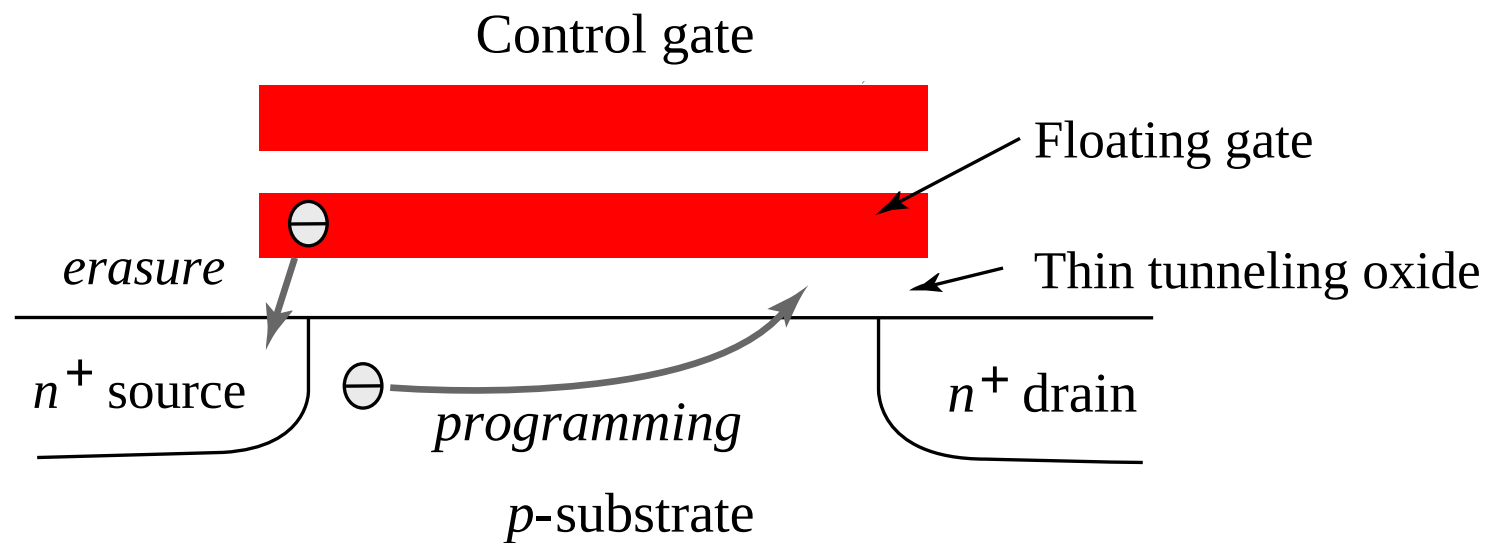


**Absolute threshold control
is hard**

**Unprogrammed transistor
might be depletion**

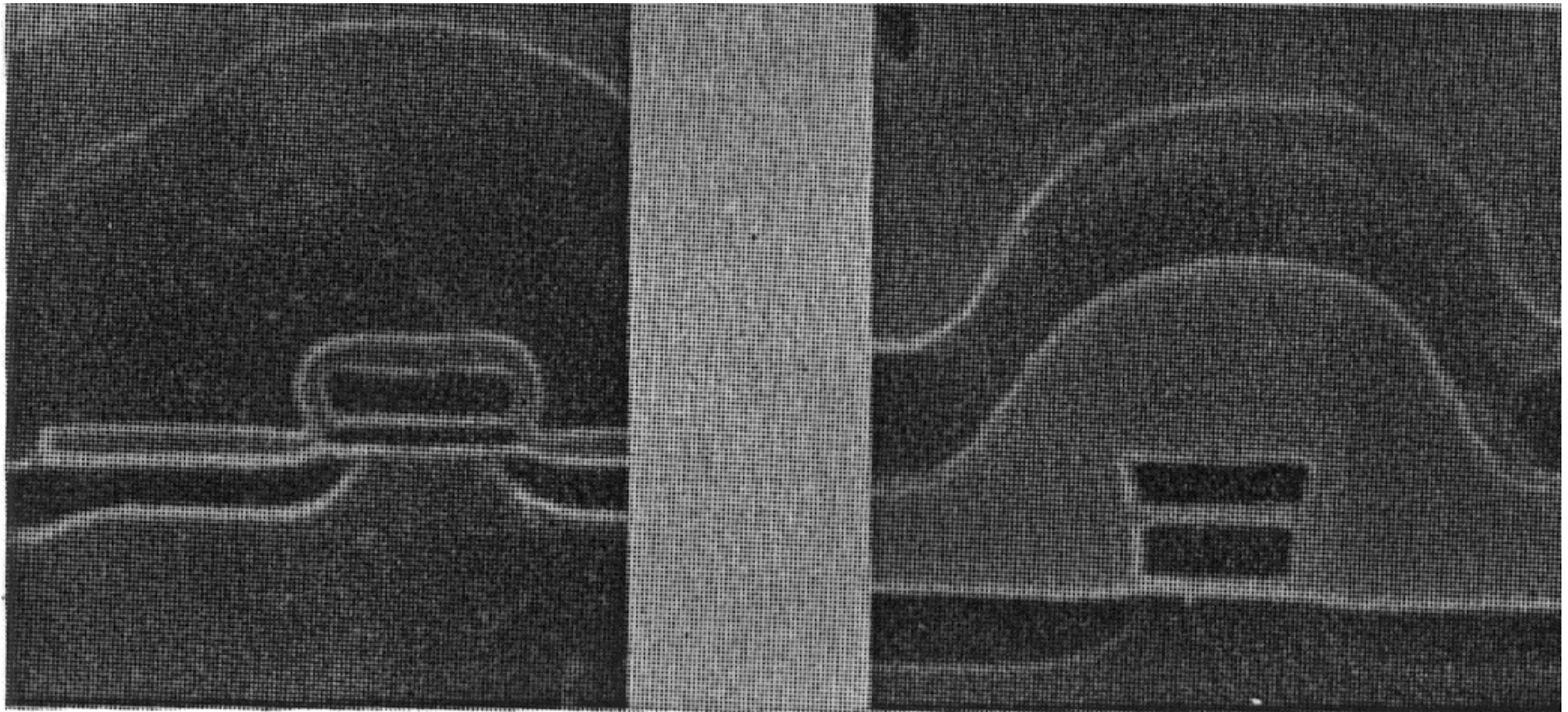
⇒ **2 transistor cell**

Flash EEPROM



Many other options ...

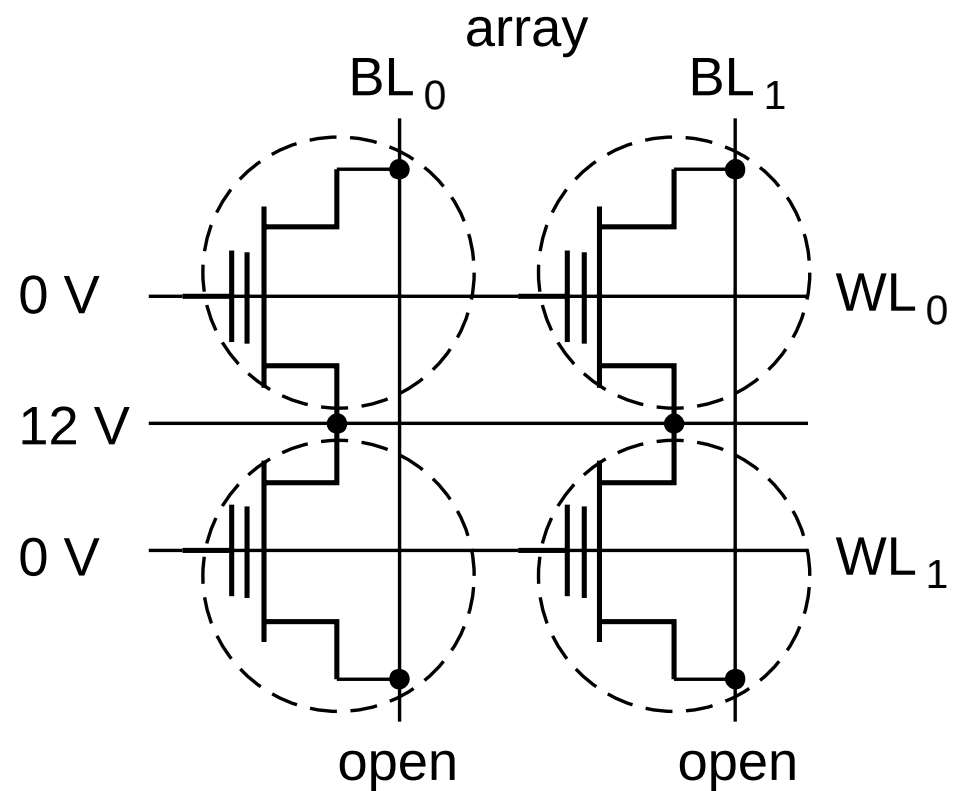
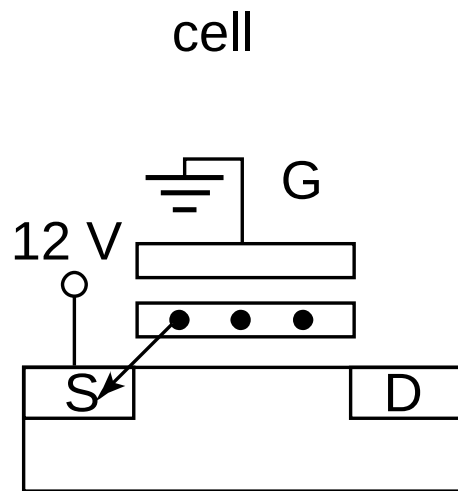
Cross-sections of NVM cells



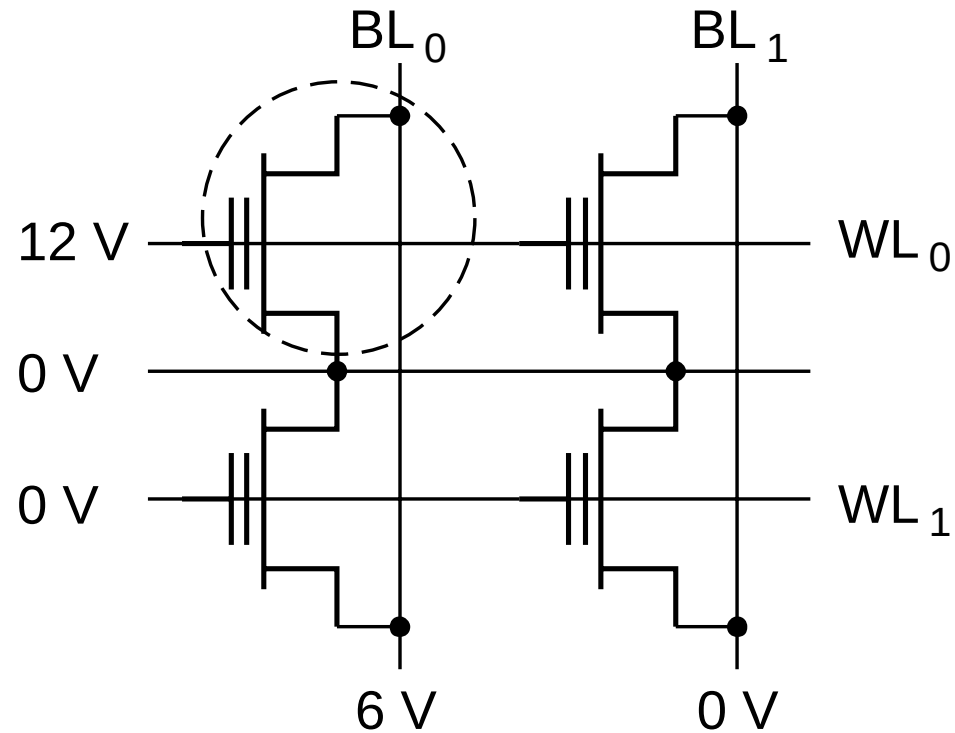
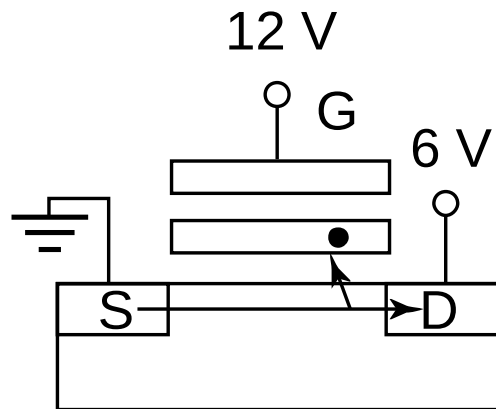
Flash

EPROM

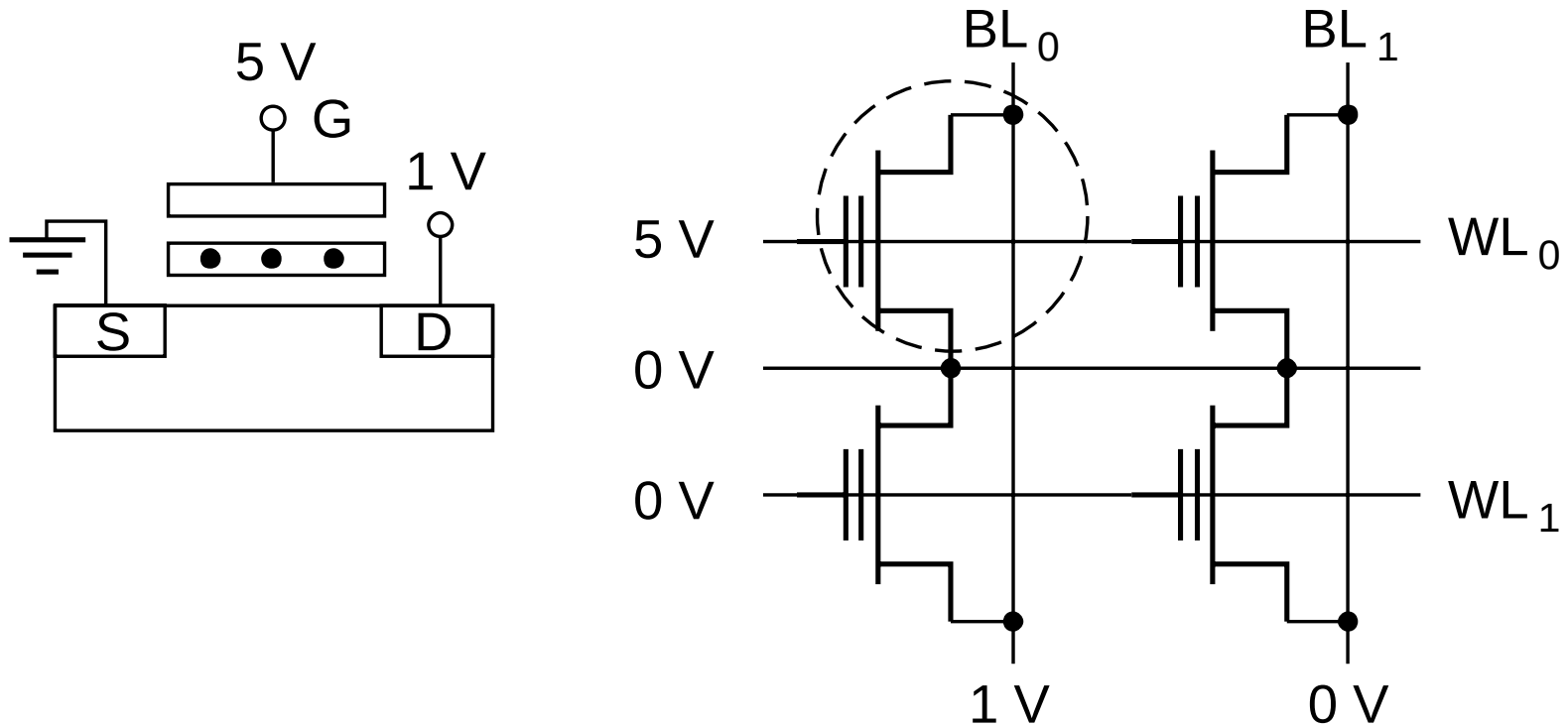
Basic Operations in a NOR Flash Memory— Erase



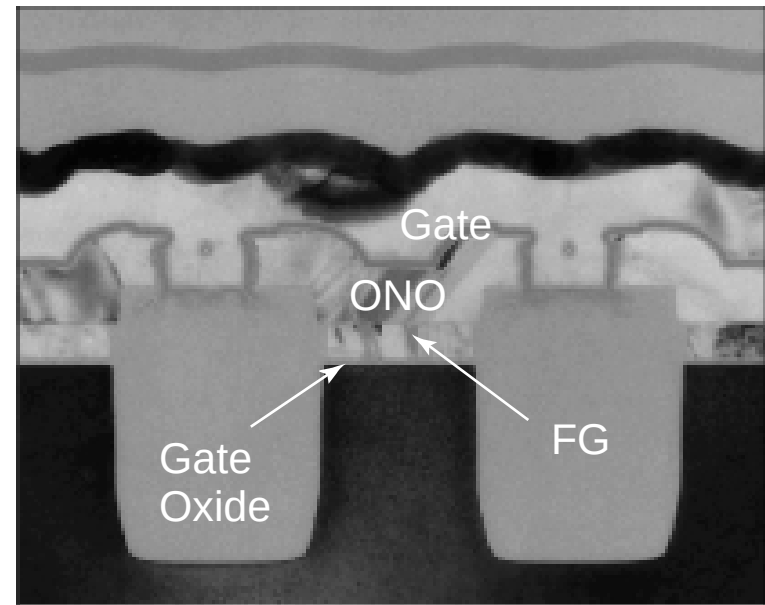
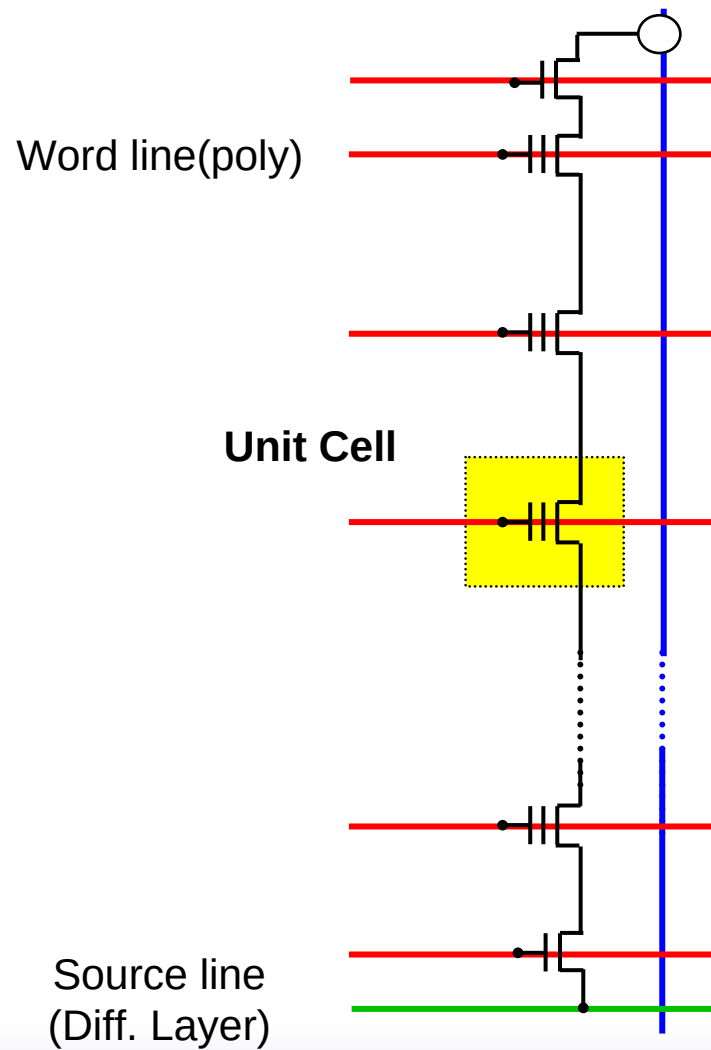
Basic Operations in a NOR Flash Memory— Write



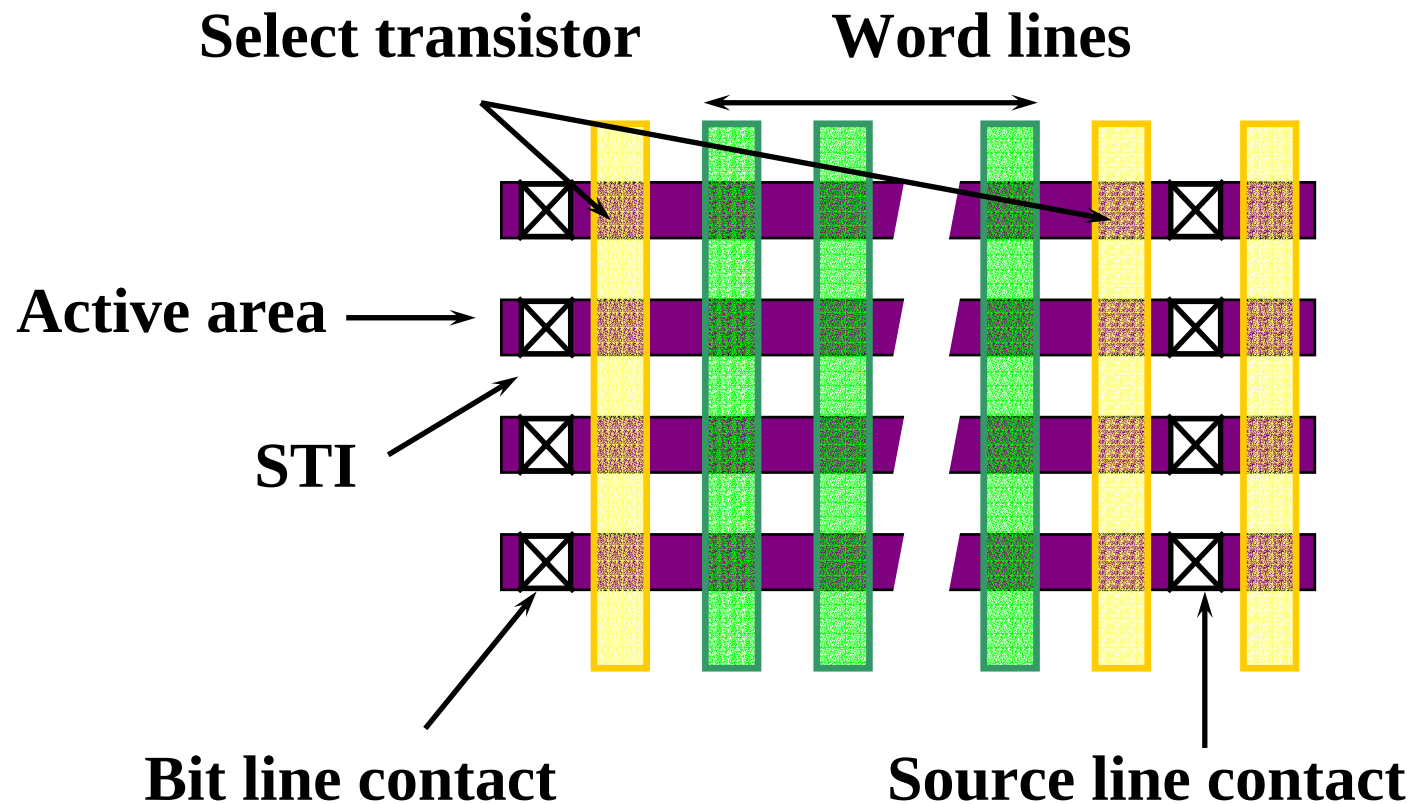
Basic Operations in a NOR Flash Memory— Read



NAND Flash Memory



NAND Flash Memory



Characteristics of State-of-the-art NVM

Table 12-1 Comparison between nonvolatile memories ([Itoh01]).

$V_{DD} = 3.3$ or 5 V; $V_{PP} = 12$ or 12.5 V.

	Cell— Nr. of Transistors	Cell Area (ratio wrt EPROM)	Mechanism		External Power Supply		Program/ Erase Cycles
			Erase	Write	Write	Read	
MASK ROM	1 T (NAND)	0.35–5	—	—	—	V_{DD}	0
EPROM	1 T	1	UV Exposure	Hot electrons	V_{PP}	V_{DD}	~100
EEPROM	2 T	3–5	FN Tunneling	FN Tunneling	V_{PP} (int)	V_{DD}	10^4 – 10^5
Flash Memory	1 T	1–2	FN Tunneling	Hot electrons	V_{PP}	V_{DD}	10^4 – 10^5
			FN Tunneling	FN Tunneling	V_{PP} (int)	V_{DD}	10^4 – 10^5